

Reconstructed-energy background spectra for a Quantum-Parity-Detector germanium wafer: reactor CEvNS, cosmic muons, and environmental gammas

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Coherent elastic neutrino-nucleus scattering (CEvNS) from a nuclear reactor produces germanium nuclear recoils of only tens to hundreds of eV, at the threshold of emerging quantum sensors. We present an end-to-end forward model of the reconstructed-energy differential rate spectra measured by a thin (4 in. $\times$ 4 in. $\times$ 2 mm, $\sim$ 110 g) single-sided germanium wafer read out by Quantum Parity Detectors (QPDs), for two quasiparticle-trapping designs (Ta $\rightarrow$ Al and Al $\rightarrow$ Hf). On a single unified phonon energy scale with no ionization quenching, we compute the deposited-energy spectra of the three dominant channels—reactor CEvNS, sea-level cosmic muons, and the environmental-gamma Compton continuum—and fold them through a Monte-Carlo QPD response matrix $R(E_{\text{rec}} | E_{\text{dep}})$ that includes the bandwidth-limited (25 kHz) censoring of the tunneling readout. The reactor flux, cross section, and background rates are anchored to published values: the CEvNS rate reproduces the Billard *et al.* (2017) germanium benchmark to within 2.4%, the integral muon rate matches the PDG sea-level flux within $\sim$ 30%, and the Compton edges fall at the Klein-Nishina energies. The resulting reconstructed-energy spectra show that the reactor-CEvNS signal reconstructs to tens of eV—the lowest-energy region and the flagship signal band—while cosmic-muon deposits saturate the readout and pile up into a tens-of-keV feature that is an instrumental artifact of the bandwidth ceiling rather than a physical spectral line. We report the linear-to-saturated crossover deposit energy per design and delimit the saturation region. We treat the saturated-regime response, the site-dependent environmental-gamma normalization, and the sub-1.8 MeV reactor flux as explicit, quantified modeling uncertainties. This forward model establishes the reconstructed-energy regime in which a QPD reactor-CEvNS search must operate.

I. INTRODUCTION

Coherent elastic neutrino-nucleus scattering (CEvNS) offers a compact, Standard-Model-anchored probe of reactor antineutrinos, but the nuclear recoils it produces in germanium at reactor energies are only tens to hundreds of eV, below the reach of conventional ionization or phonon detectors. First predicted by Freedman [1] and proposed as a neutrino-detection channel by Drukier and Stodolsky [2], CEvNS was observed only in 2017 by the COHERENT experiment at a stopped-pion source [3], which fixed the cross-section normalization against the electroweak Standard Model. Reactor sources move the target recoil energies an order of magnitude lower, into the tens-of-eV regime, in exchange for an enormous $\bar{\nu}_e$ flux. A dedicated effort has followed—CONUS+ [4], Dresden-II [5], and NUCLEUS [6]—each pushing the deposited-energy threshold toward and below 100 eV, where the germanium reactor-CEvNS rate is benchmarked by the calculation of Billard *et al.* [7]. The central experimental obstacle is not statistics but threshold: recovering the reactor-CEvNS spectrum demands sensitivity to nuclear recoils that terminate at a few tens of eV.

Quantum Parity Detectors (QPDs) are a superconducting readout concept aimed squarely at this regime [8]. A QPD infers a deposited energy from

the population of quasiparticles it liberates in a superconducting sensor, sensed through the parity-dependent tunneling of individual quasiparticles across a Josephson junction. Because each tunneling event is counted, the scheme offers a path to sub-keV and even eV-scale thresholds on a fieldless phonon calorimeter. It carries, however, an intrinsic limitation: the tunneling signal is bandwidth-limited, and the readout response *saturates* once the instantaneous tunneling rate exceeds the resolving ceiling—here $\sim$ 25 kHz, set by a $\tau_d \simeq 40 \mu\text{s}$ dead time. A large deposit drives the burst rate far above this ceiling, so the number of *recorded* tunneling events, and hence the inferred energy, no longer tracks the energy actually deposited. The decisive observable for such a detector is therefore not the deposited energy but the *reconstructed* energy E_{rec} that survives the bandwidth-limited counting chain. A background that is innocuous in deposited energy can, after reconstruction, land squarely on top of an eV-scale signal.

This paper computes that reconstructed-energy observable end to end. We build a forward model that maps the deposited-energy spectrum of the reactor-CEvNS signal and of its two dominant surface backgrounds—through-going cosmic-ray muons and the radiogenic environmental-gamma continuum—into reconstructed-energy differential rate spectra dR/dE_{rec} for a thin single-sided germanium wafer read out by QPDs. We treat two quasiparticle-trapping designs, Ta $\rightarrow$ Al and Al $\rightarrow$ Hf [8], and we place all three channels on a single unified phonon energy scale with no ionization quenching, as

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befits a fieldless calorimeter, so that signal and backgrounds are directly co-added and folded through a common response. Every rate is anchored to a published value rather than assumed: the deposited CEvNS rate reproduces the Billard germanium benchmark to within 2.4%, the cosmic-muon flux matches the sea-level value tabulated by the Particle Data Group to within its $\sim 30\%$ inter-experiment spread [9], and the environmental-gamma Compton edges are fixed exactly by scattering kinematics. The response chain that carries these deposited spectra into E_{rec} —the sensor-sharing partition, the tunneling-burst statistics, and the non-paralyzable censoring at the 25 kHz ceiling—is the methodological core of the work.

The headline finding is a qualitative separation on the reconstructed axis. The reactor-CEvNS signal, whose recoils lie entirely below the saturation onset, reconstructs *linearly* to tens of eV, peaking near 42 eV in E_{rec} with $\sim 85\%$ of its counts below 100 eV. The cosmic-muon channel, by contrast, deposits $\sim 1.5\text{--}197$ MeV and drives the readout deep into saturation, so its entire deposited spectrum collapses onto a single tens-of-keV plateau (≈ 18.8 keV for Ta $\rightarrow$ Al, ≈ 15.0 keV for Al $\rightarrow$ Hf). We stress at the outset what this separation is and is not. It is a stage-1 forward-model, or feasibility, calculation, not a sensitivity or discovery claim, and we quote no detection significance. The muon pile-up is an *instrumental artifact* of the modeled bandwidth ceiling, not a physical spectral line, and the shape of the saturated-regime response has no published anchor; the absolute environmental-gamma normalization is a site-dependent estimate good to a factor of ~ 2 ; and the crossover between the linear and saturated regimes rides on two low-confidence sensor-sharing parameters, so we report it as a band spanning roughly 50 eV to 13 keV rather than a single number. These modeling uncertainties are quantified channel by channel throughout, so that the separation stands on stated assumptions rather than hidden ones.

The remainder of the paper follows the pipeline in order. Section II fixes the detector geometry, the unified phonon energy scale, and the per-sensor tunneling chain. Section III builds the deposited reactor-CEvNS signal from an absolute antineutrino flux and the coherent cross section, and anchors it to the Billard benchmark. Section IV computes the deposited cosmic-muon and environmental-gamma spectra. Section V constructs the bandwidth-limited response matrix $R(E_{\text{rec}} | E_{\text{dep}})$ that is the heart of the forward model. Section VI folds the three channels through it to obtain the reconstructed spectra of Fig. 1, and Secs. VII and VIII discuss the implications and limits of the result.

II. DETECTOR MODEL AND ENERGY SCALE

The detector is a single-sided germanium wafer, 4 in. $\times$ 4 in. $\times$ 2 mm, of density $\rho = 5.323$ g cm $^{-3}$, giving a volume of ~ 20.6 cm 3 and a mass of ~ 110 g. At

the natural-abundance molar mass this corresponds to 8.29×10^{24} germanium atoms per kilogram; we quote all differential rates per unit target mass (counts kg $^{-1}$ day $^{-1}$ keV $^{-1}$) so they can be compared directly with the reactor-CEvNS benchmark literature, while the physical target for absolute rates is the 110 g wafer. One face carries $\sim 10,300$ Quantum Parity Detector (QPD) sensors at a 1 mm pitch, each a superconducting quasiparticle trap read out through a charge-parity junction [8]. We consider the two trapping designs of that work, Ta $\rightarrow$ Al and Al $\rightarrow$ Hf, whose device parameters are taken verbatim from its Table II and summarized in Table I. The two designs differ only in the trap material and its junction parameters; they share the geometry, the energy scale, and the reconstruction chain developed below.

Unified phonon energy scale. A fieldless phonon calorimeter has no ionization-collection channel and no Luke–Neganov gain, so the full recoil energy of a deposit thermalizes into the phonon system regardless of whether the primary recoil is nuclear (CEvNS) or electronic (cosmic-muon ionization, Compton). We therefore place every channel on a single unified phonon energy scale with *no* ionization quenching: a nuclear recoil and an electron recoil of equal deposited energy E_{dep} land at the same point on the observable axis. Applying a Lindhard or ionization-yield factor here would suppress the CEvNS rate by $\sim 5\text{--}7\times$ and would be physically incorrect for this sensor [1]; we consequently never mix keV $_{\text{ee}}$ and keV $_{\text{nr}}$ scales. The phonon energy is $E_{\text{ph}} = E_{\text{dep}} - E_{\text{stored}}$, where the defect (Frenkel-pair) storage E_{stored} is a few-percent nuclear-recoil-only correction and is zero for muon and Compton deposits; we take $E_{\text{ph}} \simeq E_{\text{dep}}$ throughout.

Deposit-to-signal chain. The phonon energy is partitioned among the sensors (the localized-plus-diffuse partition, which sets the per-sensor share E_s , is specified in Sec. V). A per-sensor share E_s breaks Cooper pairs to yield a trapped-quasiparticle count N_{qp} , a density n_{qp} over the effective trap volume V_{tr} , and a plateau tunneling rate Γ_{in} set by the per-design tunneling constant K :

$$\begin{aligned} N_{\text{qp}} &= \varepsilon \frac{E_s}{\Delta_{\text{tr}}}, & n_{\text{qp}} &= \frac{N_{\text{qp}}}{V_{\text{tr}}}, \\ \Gamma_{\text{in}} &= K n_{\text{qp}}, & \Gamma_{\text{in}}^{\text{pk}} &= p \Gamma_{\text{in}}, \end{aligned} \quad (1)$$

where Δ_{tr} is the trap superconducting gap and ε the deposited-to-signal efficiency (below). The steady rate $\Gamma_{\text{in}} = K n_{\text{qp}}$ is dimensionally consistent (K in Hz μm^3 , n_{qp} in μm^{-3}), and $N_{\text{qp}} = \varepsilon E_s / \Delta_{\text{tr}}$ is a pure count. The instantaneous burst peaks above its plateau by a two-exponential pulse peak factor p (from the injection/recombination profile of Ref. [8], Eq. 4), giving the peak rate $\Gamma_{\text{in}}^{\text{pk}}$ that governs saturation.

Efficiency. We impose $\varepsilon \approx 0.5$ as a forward-model definition lumping phonon collection, pair-breaking, and trapping, so that in the unsaturated regime the reconstructed energy is $E_{\text{rec}} \approx \varepsilon E_{\text{dep}} = 0.5 E_{\text{dep}}$. This is a definitional baseline, not a derived value; we carry a

TABLE I. Trap device parameters for the two designs, from Table II of Ref. [8]. Δ_{tr} is the trap gap, V_{tr} the effective trap volume, K the tunneling constant ($\Gamma_{\text{in}} = Kn_{\text{qp}}$), and p the two-exponential peak factor.

Parameter	Ta→Al	Al→Hf
Trap material	Al	Hf
Δ_{tr} (μeV)	190	40
V_{tr} (μm^3)	100	1000
K ($\text{kHz } \mu\text{m}^3$)	3	20
p	0.25	0.13

design-dependent ± 10 – 20% band on it and adopt the same central value for both designs. It is distinct from, and should not be conflated with, the paper’s physical efficiency estimate $\eta_{\text{ce}} \approx 0.3$, which we retain only as an independent cross-reference.

Bandwidth-limited readout and saturation. The tunneling readout is sampled at 50 kHz (interval 20 μs), which by the Nyquist criterion resolves tunneling transients up to 25 kHz, i.e. a resolving (dead) time $\tau_d = 40 \mu\text{s}$. A deposit saturates the readout when its peak tunneling rate exceeds this ceiling,

$$\Gamma_{\text{in}}^{\text{pk}} > \Gamma_{\text{max}} \equiv \frac{1}{\tau_d} = 25 \text{ kHz}, \quad \tau_d = 40 \mu\text{s}. \quad (2)$$

Because Δ_{tr} , V_{tr} , K , and p differ between designs, the two designs reach this ceiling at different deposit energies; the Al→Hf design saturates first (its plateau Γ_{in} per unit deposit is $\sim 3.2\times$ that of Ta→Al). The per-design crossover deposit energy and the whole-array plateau are quoted in Sec. V. The mapping from the true tunneling rate Γ to the observed count rate m is a dead-time censoring model, and the choice of model is not fixed by electronics reasoning alone; both the non-paralyzable and paralyzable idealizations are defensible:

$$\begin{aligned} m &= \frac{\Gamma}{1 + \Gamma\tau_d} \quad (\text{non-paralyzable}), \\ m &= \Gamma e^{-\Gamma\tau_d} \quad (\text{paralyzable}). \end{aligned} \quad (3)$$

The non-paralyzable form is monotone and saturates at $m \rightarrow 1/\tau_d = 25 \text{ kHz}$; the paralyzable form is non-monotone, peaking at $\Gamma = 1/\tau_d$ and rolling over thereafter. This was an open modeling choice, closed by decision: we adopt *non-paralyzable* censoring as the canonical convention for all results and retain the paralyzable variant only as a sensitivity. Equations (1)–(3) fix the setup on which the flux and background channels (Secs. III and IV) and the reconstruction estimator (Sec. V) are built.

III. REACTOR ANTINEUTRINO FLUX AND CEVNS RATE

The reactor antineutrino signal enters our pipeline as a deposited nuclear-recoil spectrum dR/dT , where $T \equiv E_{\text{nr}}$ is the germanium recoil kinetic energy on the unified phonon scale of Sec. II. We build it in three steps: an absolute antineutrino flux $\Phi(E_\nu)$, the coherent-scattering cross section $d\sigma/dT$, and their fold over the five natural germanium isotopes. All rates in this section are on the *deposited* energy T ; the mapping to reconstructed energy E_{rec} is deferred to Secs. V and VI.

A. Reactor antineutrino flux

We take the emitted per-fission spectra $S_i(E_\nu)$ from the Huber conversion fits [10] for ^{235}U , ^{239}Pu , and ^{241}Pu and from the Mueller fit [11] for ^{238}U above 2 MeV, with coefficients taken from the primary arXiv e-prints; our reconstructed ^{235}U spectrum agrees with the published Huber tabulation to 0.73% at 3 MeV and 0.13% at 5 MeV. Below 1.8 MeV, where the conversion fits do not extend, we seam-match (continuous through the derivative) a summation-style continuation plus a $^{238}\text{U}(n, \gamma)$ component normalized to 0.6 $\bar{\nu}_e$ per fission. The absolute flux at the detector is

$$\begin{aligned} \Phi(E_\nu) &= \frac{R_f}{4\pi d^2} \left[\sum_i f_i S_i(E_\nu) + S_{n\gamma}(E_\nu) \right], \\ R_f &= \frac{P_{\text{th}}}{\langle E_f \rangle}, \end{aligned} \quad (4)$$

where R_f is the fission rate, $P_{\text{th}} = 3 \text{ GW}_{\text{th}}$ is the *thermal* reactor power, $\langle E_f \rangle = 205.8 \text{ MeV}$ is the fission-fraction-weighted effective thermal energy per fission (not the total Q -value, which double-counts the escaping neutrinos), f_i are the fission fractions, and $d = 25 \text{ m}$ is the standoff, treated as a point source. This gives $R_f = 9.10 \times 10^{19} \text{ fissions s}^{-1}$ and an integrated flux $\int \Phi dE_\nu = 7.5 \times 10^{12} \bar{\nu}_e \text{ cm}^{-2}\text{s}^{-1}$. The corresponding emission of $1.96 \times 10^{20} \bar{\nu}_e \text{ s}^{-1}\text{GW}_{\text{th}}^{-1}$ reproduces the Hayes–Vogel reactor-accounting value $\sim 2 \times 10^{20}$ [12]. The sub-1.8 MeV shape is a model placeholder consistent with, but not directly digitized from, Kopeikin [13]; we carry it with a wide below-2-MeV uncertainty band and quantify its downstream impact at the end of this section.

B. Coherent cross section

For each isotope i we use the Freedman coherent cross section [1, 2],

$$\frac{d\sigma_i}{dT} = \frac{G_F^2 M_i}{4\pi} Q_{W,i}^2 \left(1 - \frac{M_i T}{2E_\nu^2} \right) F_i^2(q), \quad (5)$$

with the $/4\pi$ prefactor (the $/8\pi$ form is a factor-of-two error), G_F the Fermi constant, M_i the nuclear mass, and weak charge $Q_{W,i} = N_i - (1 - 4\sin^2\theta_W)Z$ evaluated at the low-energy value $\sin^2\theta_W = 0.2387$. Because $1 - 4\sin^2\theta_W = 0.045$, the proton coupling nearly vanishes and the rate scales as N^2 to $\sim 0.5\%$. The momentum transfer is $q = \sqrt{2M_i T}$ and $F_i(q)$ is the Lewin–Smith Helm form factor [14, 15], which stays above 0.998 at reactor recoil energies on germanium. The cross section is evaluated in natural units and restored to cm^2 with $(\hbar c)^2 = 3.894 \times 10^{-28} \text{ GeV}^2 \text{cm}^2$; dropping this factor is the single most common normalization error in reactor CEvNS. Equation (5) integrates to the closed form $\sigma_{\text{tot}} = G_F^2 Q_W^2 E_\nu^2 / 4\pi$ to better than 6×10^{-8} , and yields $\sigma(^{72}\text{Ge}, 4 \text{ MeV}) = 1.0026 \times 10^{-40} \text{ cm}^2$, within 0.26% of the first-principles anchor. The Helm parameters and the per-isotope germanium table are collected in Sec. A.

C. Deposited-energy rate

The differential rate follows from folding Eq. (4) against Eq. (5) over the five stable germanium isotopes, retaining each isotope’s own weak charge, nuclear mass, and kinematic threshold rather than a lumped- A approximation:

$$\frac{dR}{dT} = N_T \sum_i x_i \int_{E_\nu^{\min}(T)}^\infty \Phi(E_\nu) \frac{d\sigma_i}{dT} dE_\nu, \quad (6)$$

$$E_\nu^{\min}(T) = \sqrt{\frac{1}{2} M_i T},$$

where $N_T = 8.29 \times 10^{24} \text{ Ge atoms kg}^{-1}$ is the target density, x_i the natural isotopic abundances (neutron numbers $N_i = 38, 40, 41, 42, 44$), and $E_\nu^{\min}(T) = \sqrt{M_i T/2}$ is the minimum antineutrino energy that produces a recoil T . Integrating Eq. (6) gives the deposited spectrum dR/dT shown in Fig. 4, with 67.8 counts $\text{kg}^{-1} \text{day}^{-1}$ above a 50 eV_{nr} deposited threshold at 3 GW_{th} and 25 m. Propagating the split flux band through Eq. (6) yields a 1σ uncertainty of 3.4–10% on the deposited rate, dominated at the lowest recoils by the sub-1.8 MeV placeholder: it contributes 18–34% of the rate below 95 eV_{nr}, corresponding to a ~ 6 –10% band there, while the well-anchored $> 1.8 \text{ MeV}$ flux controls the rate at every higher recoil.

D. Absolute-normalization benchmark

The decisive external check is a reproduction of the Billard *et al.* germanium reactor-CEvNS rates [7]. Their Table 1 is quoted for a 8.54 GW_{th} source at $\sim 400 \text{ m}$, so we rescale a flux variant built under their assumptions (Huber–Mueller held constant below 2 MeV, no n -capture) to that configuration by the geometry-and-

power factor

$$k = \frac{P_B}{P_{\text{th}}} \left(\frac{d}{d_B} \right)^2 = 0.01112, \quad (7)$$

with $P_B = 8.54 \text{ GW}_{\text{th}}$ and $d_B \simeq 400 \text{ m}$; both powers are thermal, so only the $1/d^2$ geometry survives in the ratio. Single-source and explicit two-core evaluations of Eq. (7) agree to 0.25%. Folding with k reproduces the Billard rates 0.76/0.51/0.26 counts $\text{kg}^{-1} \text{day}^{-1}$ above 50/100/200 eV_{nr} as 0.742/0.501/0.257, within 2.4%. Omitting k overshoots by exactly $1/k = 89.9\times$ in every bin, confirming that the residual mismatch is purely the geometry-and-power normalization and not the spectral shape or the cross section. As a second, coarser anchor, rescaling both our flagship prediction and the Billard rate to the CONUS+ configuration [4] agrees to a factor 0.99, though this comparison is at the ionization (eV_{ee}) scale and requires a quenching model to relate to our deposited phonon scale, so we treat it as a rate-scale cross-check only. Together these anchors fix the absolute normalization of the deposited CEvNS signal that Secs. V and VI carry through the QPD response.

IV. COSMIC-MUON AND ENVIRONMENTAL-GAMMA BACKGROUNDS

Two irreducible ambient radiation fields dominate the deposited-energy budget of an unshielded surface wafer: through-going cosmic-ray muons and the radiogenic environmental-gamma continuum. We compute each deposited spectrum dR/dE_{dep} on the unified phonon scale (no ionization quenching; both channels are electron recoils), and we anchor each to a published benchmark so that the downstream reconstructed spectra rest on validated inputs rather than assumed shapes. Neither channel involves new physics; the care lies entirely in the thin-wafer geometry, the straggling statistics, and the single-scatter kinematics.

A. Cosmic muons

We sample the sea-level muon intensity from the modified-Gaisser (“Gaisser–Guan”) parametrization [16, 17], which incorporates the low-energy/large-angle correction that plain Gaisser lacks,

$$\frac{dI}{dE_\mu d\Omega} = 0.14 \left(\frac{E_\mu}{\text{GeV}} \right)^{-2.7} \times \left[\frac{1}{1 + \frac{1.1 E_\mu \cos \theta^*}{115 \text{ GeV}}} + \frac{0.054}{1 + \frac{1.1 E_\mu \cos \theta^*}{850 \text{ GeV}}} \right], \quad (8)$$

in $\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1} \text{GeV}^{-1}$, where E_μ is the muon energy and the effective zenith angle $\cos \theta^* = [(\cos^2 \theta + P_1^2 + P_2 \cos^{P_3} \theta + P_4 \cos^{P_5} \theta) / (1 + P_1^2 + P_2 + P_4)]^{1/2}$ softens the

horizontal divergence of the plain $\sec \theta$ scaling ($P_1 - P_5 = 0.102573, -0.068287, 0.958633, 0.0407253, 0.817285$).

Each muon piercing the wafer traverses a chord ℓ fixed by the ray-box intersection of its straight trajectory with the $10.16 \times 10.16 \times 0.20$ cm slab. The chord ranges from the 0.20 cm vertical thickness to the 14.37 cm space diagonal, with the isotropic-flux mean pinned by Cauchy's theorem to $\langle \ell \rangle = 4V/S = 0.385$ cm. The deposited spectrum is the flux-weighted, projected-area chord distribution folded with the straggling density,

$$\frac{dR}{dE_{\text{dep}}} = \frac{1}{m} \int d\Omega (\hat{n} \cdot \hat{\Omega})_+ A \int dE_\mu \frac{dI}{dE_\mu d\Omega} \times f_L(E_{\text{dep}}; \Delta_p(\ell, E_\mu), \xi(\ell)), \quad (9)$$

where m is the wafer mass, A the entry-face area, $(\hat{n} \cdot \hat{\Omega})_+$ the inward surface projection, and f_L the Landau-Vavilov density whose mode sits at the *most-probable value* (MPV) Δ_p . Using the mean deposit here, or the Moyal approximation, would misplace the peak and suppress the high-energy tail; we use neither. With mass thickness $x = \rho\ell$, the width and MPV follow the standard forms

$$\xi = \frac{K Z x}{2 A \beta^2}, \quad \Delta_p = \xi \left[\ln \frac{2m_e c^2 \beta^2 \gamma^2}{I} + \ln \frac{\xi}{I} + j - \beta^2 - \delta(\beta\gamma) \right], \quad (10)$$

with $K = 0.307 \text{ MeV mol}^{-1} \text{ cm}^2$, $Z/A = 0.4406$ for Ge, mean excitation $I \approx 350 \text{ eV}$, $j = 0.200$, and $\delta(\beta\gamma)$ the density-effect correction. For a vertically incident minimum-ionizing muon ($x = 1.065 \text{ g cm}^{-2}$) we obtain $\xi = 0.072 \text{ MeV}$, matching the PDG value [9], and $\Delta_p = 1.23 \text{ MeV}$ — strictly below the mean deposit $\langle \Delta \rangle = 1.46 \text{ MeV}$. This ordering $\Delta_p < \langle \Delta \rangle$ is the signature of the strongly right-skewed Landau distribution and is the single most consequential modeling choice in this channel.

Rare near-horizontal trajectories traverse up to the full 14.37 cm diagonal and deposit up to $\sim 197 \text{ MeV}$; we importance-sample these long chords so this high-deposit tail is statistically resolved. It is the tail — not the $\sim \text{MeV}$ bulk — that drives the bandwidth-limited readout into saturation, as analyzed in Sec. V. Integrating Eq. (9) over flux and projected area gives a total muon rate through the wafer of 1.366 Hz. This is consistent with the PDG sea-level expectation ($\sim 1 \text{ muon cm}^{-2} \text{ min}^{-1}$, $I_v \approx 70 \text{ m}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$) to within $\sim 20\%$, well inside the VALD-02 validation tolerance of 30%; the residual reflects the known 30–35% inter-experiment normalization spread of the Gaisser–Guan flux and is the weakest anchor of this otherwise textbook-grade channel.

B. Environmental gammas

The wafer sits in a radiogenic gamma field sourced by ^{40}K and the ^{238}U and ^{232}Th decay chains. The line energies are fixed nuclear data; we take the absolute per-line

fluxes from the LABChico measured survey [18] (for example, the ^{40}K 1461 keV line at $0.036 \text{ cm}^{-2} \text{ s}^{-1}$), with the line list following the standard low-background compilation [19]. Because the 2 mm wafer is optically thin to MeV photons — the interaction probability is $\mu\ell \approx 0.117$ per mean chord at 1 MeV, so the double-scatter fraction is only $\sim 1.4\%$ — each photon Compton-scatters at most once and the scattered photon escapes. The deposit is therefore the recoil-electron kinetic energy $T_e = E_\gamma - E'$, a *continuum*, and not a full-energy photopeak.

We sample the scattering angle θ from the bound-electron incoherent cross section,

$$\frac{d\sigma}{d\Omega} = \frac{r_e^2}{2} \left(\frac{E'}{E_\gamma} \right)^2 \left(\frac{E'}{E_\gamma} + \frac{E_\gamma}{E'} - \sin^2 \theta \right) S(x, Z), \quad (11)$$

$$E' = \frac{E_\gamma}{1 + \alpha(1 - \cos \theta)},$$

with classical electron radius r_e , incident photon energy E_γ , scattered energy E' , and $\alpha = E_\gamma/m_e c^2$. The Klein–Nishina factor [20] sets the continuum shape and the free-electron edge; the Hubbell incoherent scattering function [21] $S(x, Z=32)$, evaluated at the momentum-transfer variable

$$x = \frac{E_\gamma [\text{keV}] \sin(\theta/2)}{12.39842} \text{ \AA}^{-1}, \quad (12)$$

imposes electron binding: since $S(x \rightarrow 0) \rightarrow 0$, it suppresses near-forward (low-recoil) scattering, rolling the deposited continuum off near 30–50 eV and removing the unphysical free-Klein–Nishina low-energy plateau, while $S(x \rightarrow \infty) \rightarrow Z$ leaves the bulk Compton continuum unchanged. The recoil deposit for each line terminates at the Compton edge,

$$E_{\text{edge}} = \frac{2E_\gamma^2}{m_e c^2 + 2E_\gamma}, \quad (13)$$

which places the ^{40}K , ^{214}Bi , and ^{208}Tl edges at 1243, 1541, and 2382 keV respectively. These edge positions are the sharpest, most model-independent features of the channel: they are fixed by Compton kinematics alone and are reproduced by the angle-sampled maxima to better than 0.5 keV.

Summed over lines, the single-scatter interaction rate is 0.267 Hz, within 2% of the independent $\Phi \sigma_{\text{KN}} N_e$ estimate (0.273 Hz, ratio 0.98), where Φ is the integral gamma flux, σ_{KN} the Klein–Nishina cross section, and N_e the wafer electron number. We are explicit about the split in confidence: the edge positions are exact (HIGH confidence), whereas the absolute normalization inherits the site-dependent gamma flux and is a representative estimate good to a factor of ~ 2 (MEDIUM confidence). The ^{238}U -chain flux in particular is set by an assumed chain balance rather than a directly measured line intensity, and a different deployment site would rescale the overall rate — but not the edge structure — accordingly.

Both deposited spectra are computed on the shared phonon energy grid, so they are directly co-added and

folded through the detector response of Sec. V to yield the reconstructed background spectra of Fig. 1. There the gamma continuum populates the sub-keV to few-keV reconstructed region, while the muon high-deposit tail is the agent of readout saturation.

V. QPD RESPONSE AND ENERGY RECONSTRUCTION

The device model of Sec. II fixed the per-sensor chain from a sensor share E_s to the peak tunneling rate $\Gamma_{\text{in}}^{\text{pk}}$ and the saturation ceiling $\Gamma_{\text{max}} = 1/\tau_d = 25$ kHz [Eqs. (1)–(3)], but deferred two ingredients to this section: how the deposited energy E_{dep} is partitioned among the $\sim 10,300$ sensors, and how a reconstructed energy E_{rec} is estimated from the resulting censored counts. We specify both here and use them to build the Monte-Carlo response matrix $R(E_{\text{rec}} | E_{\text{dep}})$ that Sec. VI folds through the deposit spectra. This is the methodological core of the pipeline, and also its principal source of model risk: the saturated-regime response has no published benchmark, and the crossover between the linear and saturated regimes rides on two low-confidence sharing parameters. We are explicit about both below.

Sensor-sharing partition. A single deposit does not illuminate the $N_{\text{sens}} \simeq 10,300$ sensors uniformly. We adopt a localized-plus-diffuse partition: a prompt fraction f_{prompt} of E_{dep} is absorbed near the impact point and spread over a spot of $\sim \pi r^2$ sensors, while the remaining $(1 - f_{\text{prompt}})$ thermalizes diffusely across the whole array. The on-spot and off-spot per-sensor shares are then

$$\begin{aligned} E_s^{\text{on}} &= \frac{f_{\text{prompt}} E_{\text{dep}}}{\pi r^2} + \frac{(1 - f_{\text{prompt}}) E_{\text{dep}}}{N_{\text{sens}}}, \\ E_s^{\text{off}} &= \frac{(1 - f_{\text{prompt}}) E_{\text{dep}}}{N_{\text{sens}}}, \end{aligned} \quad (14)$$

where r is the spot radius in sensor-pitch units. Both sides carry units of energy. The two parameters f_{prompt} (default 0.3, range 0.1–0.5) and r (default 2, range 1–5) have no direct thin-wafer QPD measurement; we treat them as exposed, scanned inputs rather than derived values, and we carry the equal-split limit $f_{\text{prompt}} \rightarrow 0$ (uniform sharing) as a required sanity bound. Because E_s^{on} is the largest per-sensor share, the on-spot sensors are the first to saturate, and f_{prompt} and r set the deposit energy at which that happens.

Count-integral estimator. On each sensor the trapped-quasiparticle population $N_{\text{qp}}(E_s)$ [Eq. (1)] drives a two-exponential tunneling burst. The quantity that controls both the burst statistics and the saturation is the expected number of *tunneling events*, $n_{\text{ev}} = \int \Gamma_{\text{in}}(t) dt = K\tau_{\text{qp}}N_{\text{qp}}/V_{\text{tr}}$, where τ_{qp} is the trapped-quasiparticle lifetime; this is distinct from, and much smaller than, the trapped count N_{qp} itself (by ~ 0.03 for Al, ~ 0.008 for Hf), and it is n_{ev} , not N_{qp} , that we feed as the Poisson mean to the exponentially-modified-Gaussian (EMG)

tunneling-burst template of Ref. [8]. Conflating the two would mis-scale the entire saturation. The template realizes the event train, which we censor with the dead-time model $m(\Gamma_{\text{in}})$ [Eq. (3)] to obtain a per-sensor observed count $N_{\text{obs},i}$; summing over the array gives the reconstructed energy

$$\begin{aligned} E_{\text{rec}} &= C N_{\text{obs}}, \quad N_{\text{obs}} = \sum_{i=1}^{N_{\text{sens}}} N_{\text{obs},i}, \\ N_{\text{obs},i} &= \int m(\Gamma_{\text{in},i}(t)) dt. \end{aligned} \quad (15)$$

The count-to-energy constant C (units eV per event) is a *single global* constant per design, fixed once on a deeply unsaturated deposit by requiring the linear slope $dE_{\text{rec}}/dE_{\text{dep}} = \varepsilon = 0.5$. It is not tuned per energy bin. Consequently, the recovery of $E_{\text{rec}} \approx 0.5 E_{\text{dep}}$ at low energy is a calibration-consistency check, not an independent validation; the genuinely predictive content of Eq. (15) is the saturated-regime behavior, to which we turn next.

Saturation and the crossover band. When the on-spot peak rate reaches the ceiling, $\Gamma_{\text{in}}^{\text{pk}} = \Gamma_{\text{max}}$ [Eq. (2)], the censored count $N_{\text{obs},i}$ stops tracking n_{ev} : under non-paralyzable censoring it *plateaus* at $\sim 1/\tau_d$ per sensor, so the summed count in Eq. (15) plateaus and E_{rec} saturates. This plateau *is* the modeled bandwidth saturation; at no point do we linearly extrapolate a rate to an energy. The retained paralyzable variant instead rolls the count over (the observed rate peaks at $\Gamma = \Gamma_{\text{max}}$ and declines), which we carry only as a sensitivity. Writing E_{onset} for the per-sensor share at which $\Gamma_{\text{in}}^{\text{pk}} = \Gamma_{\text{max}}$ (~ 1.27 eV for Ta $\rightarrow$ Al, ~ 0.77 eV for Al $\rightarrow$ Hf, from Eqs. (1)–(2)), the crossover deposit energy at which the on-spot sensor first saturates follows from inverting Eq. (14):

$$E_{\text{dep}}^{\text{cross}} = \frac{E_{\text{onset}}}{\frac{f_{\text{prompt}}}{\pi r^2} + \frac{1 - f_{\text{prompt}}}{N_{\text{sens}}}}. \quad (16)$$

Because $E_{\text{dep}}^{\text{cross}}$ inherits the f_{prompt}/r uncertainty, we report it as a band rather than a single number. At the default point it is ~ 52.9 eV (Ta $\rightarrow$ Al) and ~ 32.1 eV (Al $\rightarrow$ Hf); over the full $f_{\text{prompt}} \in [0.1, 0.5]$, $r \in [1, 5]$ scan it ranges up to ~ 13 keV and ~ 7.9 keV in the equal-split limit, with a whole-array plateau (every sensor saturated) at ~ 18.6 keV and ~ 11.3 keV. The Al $\rightarrow$ Hf design saturates before Ta $\rightarrow$ Al at every point of the scan, consistent with its lower per-sensor onset. These are the SIMU-01 design numbers forward-referenced in Sec. II; the spread of roughly three orders of magnitude is f_{prompt} -dominated and is an honest reflection of the sharing uncertainty, not a computed precision.

Response matrix. We assemble the bandwidth-limited

transfer function

$$R(E_{\text{rec}} | E_{\text{dep}}) = \frac{dP}{dE_{\text{rec}}} \Big|_{E_{\text{dep}}}, \quad (17)$$

$$\int R(E_{\text{rec}} | E_{\text{dep}}) dE_{\text{rec}} = 1,$$

the normalized conditional distribution of reconstructed energy for a fixed deposit, per design and per censoring variant. Each E_{dep} column is built by Monte-Carlo, histogramming $N_s = 5000$ forward realizations of Eq. (15) over a grid of 584 deposit energies sampled uniformly in $\log E_{\text{dep}}$ from 10.14 eV to 197 MeV (80 bins per decade), spanning the sub-keV CEvNS floor through the deep-saturated cosmic-muon tail. The $\sim \pi r^2$ on-spot sensors are realized individually with the EMG template; the $\sim 10,287$ identical off-spot sensors are drawn as one aggregate ensemble per realization rather than looped. Columns normalize to unity to 10^{-12} , and the well-populated cells converge to $\lesssim 1.8\%$ at $N_s = 5000$ with clean $1/\sqrt{N_s}$ scaling. Figure 2 shows the resulting median $E_{\text{rec}}(E_{\text{dep}})$ with its 16–84% band for both designs and both variants. At the 197 MeV muon endpoint the non-paralyzable response plateaus at ~ 34.8 keV (Ta $\rightarrow$ Al) and ~ 26.8 keV (Al $\rightarrow$ Hf)—some three to four orders of magnitude below the linear $0.5 E_{\text{dep}}$ line (≈ 98.5 MeV)—while the paralyzable variant rolls over to ~ 3.3 keV and ~ 2.6 keV. No column ramps linearly into the MeV range.

Limiting-case validation and its limits. The estimator is validated only at its two limits (VALD-04): $E_{\text{rec}} \approx 0.5 E_{\text{dep}}$ below the crossover, and a plateau once the peak tunneling rate exceeds 25 kHz, driven by the censoring model rather than by any numerical artifact (with censoring disabled, the count integral returns to exactly $0.5 E_{\text{dep}}$, so the plateau demonstrably has teeth). We state plainly that *there is no literature anchor for the shape of the response between these two limits*: the saturated-regime curve is a model prediction, not a benchmarked result. The tens-of-keV muon pile-up that this section predicts is therefore an instrumental artifact of the modeled 25 kHz ceiling—not a physical spectral feature—and should be read as “the model says,” not “the detector measures.” With this caveat in hand, Eq. (17) is the object we fold the deposited-energy spectra through in Sec. VI.

VI. RECONSTRUCTED-ENERGY SPECTRA

We now fold the three deposited-energy spectra of Secs. III and IV through the non-paralyzable response matrix $R(E_{\text{rec}} | E_{\text{dep}})$ of Sec. V to obtain the reconstructed-energy differential rate spectra dR/dE_{rec} for the two trapping designs. This is the payoff of the forward model: the reconstructed axis is what a bandwidth-limited readout actually records, and it differs qualitatively from the deposited axis wherever the tunneling

burst saturates the 25 kHz ceiling. The fold is a discrete matrix product,

$$\frac{dR}{dE_{\text{rec}}} \Big|_j = \frac{1}{\Delta E_{\text{rec},j}} \sum_i R_{ji} N_{\text{dep},i}, \quad (18)$$

$$N_{\text{dep},i} = \frac{dR}{dE_{\text{dep}}} \Big|_i \Delta E_{\text{dep},i},$$

where $N_{\text{dep},i}$ is the rate in deposited bin i , R_{ji} is the probability that a deposit in bin i is reconstructed into bin j , and $\Delta E_{\text{dep},i}$, $\Delta E_{\text{rec},j}$ are the bin widths. Because every column of R is normalized to unit sum, Eq. (18) conserves counts per channel by construction; we verify this numerically to a relative $\sim 10^{-16}$ for all three channels and both designs. The CEvNS spectrum, computed on a recoil grid in Sec. III, is first rebinned onto the shared E_{dep} grid by a piecewise log-log power-law integral that conserves counts to 9.6×10^{-5} overall and 0.12% above 50 eV_{nr}. Its sub-floor low edge (5–10.14 eV, 7.27 counts kg⁻¹ day⁻¹) lies far below the saturation onset, where the response is exactly linear; it is therefore reconstructed via the calibrated mapping $E_{\text{rec}} = 0.5 E_{\text{dep}}$ rather than discarded, and carried into the matching E_{rec} bins.

Figure 1 is the central result: dR/dE_{rec} versus reconstructed energy for CEvNS, cosmic muons, environmental gammas, and their total, for both designs; the underlying true-to-reconstructed energy mapping is shown separately in Fig. 3. Nothing below 10 eV is shown. Three qualitative features organize the spectrum.

First, the reactor-CEvNS signal—the flagship channel—reconstructs to *tens of eV*. Its differential rate peaks near 42 eV in E_{rec} , with $\sim 85\%$ of the reconstructed counts falling below 100 eV, and integrates to 109.6 counts kg⁻¹ day⁻¹. Because CEvNS recoils sit entirely below the crossover onset, they map linearly ($E_{\text{rec}} \approx 0.5 E_{\text{dep}}$, down to ~ 2.5 eV), so the flagship signal populates the very lowest end of the observable axis. This places the signal well below the sub-100 eV thresholds now being probed by reactor-CEvNS experiments [4], underscoring that recovering it demands an eV-scale reconstruction floor rather than merely a low deposited threshold.

Second, and most striking, the cosmic-muon deposits all reconstruct to a single saturation plateau. Muon energy depositions span roughly 1.5 MeV to ~ 197 MeV (Sec. IV); every one of them drives the peak tunneling rate orders of magnitude above the 25 kHz ceiling, so the count-integral estimator saturates and the entire channel piles up at $E_{\text{rec}} \approx 18.8$ keV (Ta $\rightarrow$ Al) and ≈ 15.0 keV (Al $\rightarrow$ Hf). We emphasize that this tens-of-keV feature is an *instrumental artifact of the bandwidth ceiling, not a physical spectral line*: the deposited muon spectrum is broad and structureless, and the pile-up appears only because the saturating response compresses three-to-four orders of magnitude of deposited energy onto the bounded plateau. The saturated-regime response shape has no literature anchor at any energy and

is validated by limiting cases only (Sec. V); the muon channel lies almost entirely within this regime, and the plateau energy must be read as a modeled ceiling rather than a measured quantity.

Third, the environmental-gamma Compton continuum fills the region between the two, with reconstructed peaks near 16.8 keV (Ta $\rightarrow$ Al) and 13.3 keV (Al $\rightarrow$ Hf) and an integrated rate of 2.1×10^5 counts kg $^{-1}$ day $^{-1}$. The incoherent-scattering binding suppression (Sec. IV) rolls the continuum off smoothly toward low energy near 30–50 eV E_{dep} , above the display floor, so no hard cut appears at the grid edge.

The saturating muon and Compton channels dominate the raw event rate, which raises the question of whether reconstruction remains operable. The total in-wafer event rate is 1.634 Hz (1.366 Hz muon + 0.267 Hz Compton), giving a mean inter-event time of 0.61 s, vastly longer than the ~ 40 μ s pulse. The resulting pileup occupancy is $R\tau \approx 3.3 \times 10^{-5}$ at the 20 μ s (50 kHz) sampling interval (6.5×10^{-5} at the 40 μ s resolving time). Quiescent reconstruction is therefore *not* precluded—the Phase-4/5 stop condition is not triggered—and the occupancy remains well below unity even at ten times the assumed flux.

The two designs give qualitatively similar spectra but differ in where saturation sets in. Because the Hf trap has the larger tunneling response per unit deposit, it saturates earlier and plateaus at a lower reconstructed energy than Ta $\rightarrow$ Al, shifting all three saturated features (muon pile-up, Compton peak, whole-array plateau) downward in E_{rec} ; below the crossover the two designs coincide on the linear $0.5 E_{\text{dep}}$ mapping. The shaded bands in Fig. 1 propagate the dominant systematics through Eq. (18): the CEvNS 1σ reactor-flux band (3.4% above 95 eV $_{\text{nr}}$, rising to 6.2%/9.8% at 50/20 eV $_{\text{nr}}$, with an additional ~ 6 –10% from the cited sub-1.8 MeV flux placeholder below 95 eV $_{\text{nr}}$), the environmental-gamma factor-of-two site-flux band, and the muon $\pm 30\%$ normalization band. These bands, not the plateau energies, set the honest precision of the reconstructed spectra.

VII. DISCUSSION

The reconstructed spectra of Fig. 1 are best read not as a rate prediction but as a map of *where*, on the axis a bandwidth-limited QPD readout actually records, the reactor signal and each ambient background come to rest. That map is the deliverable of this work: it tells a would-be experiment which instrumental knob is decisive before any exposure is accumulated. Its interpretation, however, is bounded by four modeling uncertainties of very different character and provenance, and an honest reading requires keeping them separate.

Saturated-regime response shape. The most consequential uncertainty is also the one with the least external support. Between the linear regime ($E_{\text{rec}} \approx 0.5 E_{\text{dep}}$) and the fully saturated plateau, the shape of $R(E_{\text{rec}} | E_{\text{dep}})$

has no published anchor at any energy; it is fixed only by its two limiting cases and by calibration consistency (Sec. V). The direct consequence is that the tens-of-keV muon pile-up of Fig. 1 is an *instrumental artifact* of the modeled 25 kHz ceiling rather than a physical spectral line. What would tighten it is unambiguous: a direct QPD saturation measurement on a thin wafer, or a full microphysical readout simulation resolving the tunneling-burst statistics through the censoring transition. Crucially, this caveat is quarantined to the channels that actually saturate—muons and the high-recoil gamma tail—because the reactor-CEvNS recoils sit entirely within the linear regime and never test the unbenchmarked shape.

Environmental-gamma normalization. The absolute Compton rate is a representative, site-dependent estimate good to a factor of ~ 2 , driven by the assumed radiogenic flux and, in particular, by the ^{238}U -chain balance rather than a measured line intensity (Sec. IV). The Compton edge positions, by contrast, are fixed by kinematics and carry high confidence. A factor-of-two shift rescales the continuum but preserves both the edge structure and the qualitative conclusion that the gamma continuum overlaps the signal region, so the *ranking* of this background is robust even though its normalization is not. Only an on-site gamma survey at the intended deployment can replace the representative estimate with a measured one.

Sub-1.8 MeV reactor flux. Below the reach of the Huber–Mueller conversion fits, the antineutrino spectrum is a cited but not directly reproduced model placeholder, carried with a wide band that translates into a ~ 6 –10% uncertainty on the reconstructed CEvNS rate below ~ 95 eV (Sec. III). This is the leading *signal-side* systematic, and it acts precisely where the reconstructed CEvNS spectrum peaks. It is, however, a band on the rate, not an ambiguity in the spectral placement: the signal remains anchored at tens of eV regardless of where in the band the true low-energy flux lies. Digitizing a per-isotope summation or CONFLUX table below 1.8 MeV, following the approach of Ref. [13], would collapse this band to the percent level.

Crossover deposit energy. The linear-to-saturated crossover rides on the two low-confidence sensor-sharing parameters f_{prompt} and r , for which no thin-wafer QPD measurement exists; we therefore report it as a band spanning ~ 50 eV to ~ 13 keV rather than a single number (Sec. V). Because the CEvNS recoils fall below even the lowest end of this band, the signal reconstructs linearly across the entire scan, so its placement is insensitive to where the true crossover sits; the uncertainty is inherited only by the saturated backgrounds, whose pile-up energies shift with the band. A dedicated thin-wafer sensor-sharing measurement would fix f_{prompt} and r and turn the band into a line.

Read together, three of the four uncertainties act on the saturated backgrounds and leave the flagship result—that reactor CEvNS reconstructs to tens of eV—as the

most robust feature of the analysis. This is the sense in which the reconstructed axis is informative despite the model risk.

The implications for a QPD reactor-CEvNS search follow directly and are mostly about detector design rather than statistics. First, *threshold placement is decisive*: the CEvNS signal populates the very lowest reconstructed energies, so recovering it demands an eV-scale reconstruction floor, not merely a low deposited threshold—a requirement that places the signal below the sub-100 eV regimes now being probed by cryogenic low-threshold detectors [6] and reactor-CEvNS experiments generally. Second, cosmic muons do *not* preclude quiescent operation: their pile-up lands at tens of keV, well *above* the CEvNS band, and the pileup occupancy is tiny ($R\tau \approx 3 \times 10^{-5}$), so muon events are sparse and spectrally displaced from the signal. A surface deployment would nonetheless benefit from muon tagging or an active veto, since the rare long-chord muon tail is the very driver of readout saturation and any misreconstruction leaks downward in energy. Third, and most limiting for a signal search, the environmental-gamma Compton continuum *overlaps* the CEvNS region, at the level of $\sim 10^3$ counts $\text{kg}^{-1}\text{day}^{-1}\text{keV}^{-1}$; unlike the muons it sits directly on top of the signal and is the dominant reducible background. Its mitigation—passive shielding and, ultimately, underground or low-background siting—is therefore the primary lever for turning this forward model into a viable search. These conclusions place the QPD concept squarely within the reactor-CEvNS program pioneered at the ionization scale by CONUS+ [4], the Dresden-II point-contact germanium result [5], and the germanium rate benchmarks of Ref. [7]; the distinction here is the unified phonon scale and the bandwidth-limited reconstruction, both inherited from the QPD device concept of Ref. [8].

We close by stating the scope plainly. This is a stage-one forward-model feasibility study: it establishes the reconstructed-energy regime in which a QPD germanium wafer would operate and identifies the design choices that regime forces. It does *not* compute a sensitivity, and it makes no claim of detectability or discovery significance. Converting this map into a sensitivity would require an exposure model, a noise-and-resolution treatment of the reconstruction floor, and—above all—the measured inputs that the four caveats above call for: a QPD saturation calibration, an on-site gamma survey, a digitized low-energy reactor flux, and a thin-wafer sensor-sharing measurement. Each is a concrete, bounded next step, and together they define the path from the present feasibility map to a quantitative reactor-CEvNS reach.

VIII. CONCLUSIONS

We have built and executed an end-to-end, published-value-anchored forward model of the reconstructed-energy background spectra for a thin single-sided ger-

manium QPD wafer. Starting from the deposited-energy spectra of the three dominant surface channels—reactor CEvNS, through-going cosmic muons, and the radiogenic environmental-gamma continuum—we placed all three on a single unified phonon energy scale with no ionization quenching, and folded them through a common bandwidth-limited response matrix $R(E_{\text{rec}}|E_{\text{dep}})$ for two quasiparticle-trapping designs, Ta $\rightarrow$ Al and Al $\rightarrow$ Hf. The output is the reconstructed-energy differential rate dR/dE_{rec} that a 25 kHz readout actually records, rather than the deposited energy that is physically inaccessible to it.

Three findings organize the result. The reactor-CEvNS signal reconstructs *linearly* to tens of eV, peaking near 42 eV in E_{rec} with $\sim 85\%$ of its counts below 100 eV: this is the flagship signal band, and it lies below the sub-100 eV thresholds now being probed by reactor-CEvNS experiments [4]. Cosmic muons, depositing ~ 1.5 –197 MeV, drive the readout deep into saturation and collapse onto a single tens-of-keV pile-up (≈ 18.8 keV for Ta $\rightarrow$ Al, ≈ 15.0 keV for Al $\rightarrow$ Hf)—an instrumental artifact of the modeled ceiling, not a physical line. The environmental-gamma Compton continuum fills the region between the two and, overlapping the signal band directly, is the dominant reducible background. The raw event rate, though dominated by the two saturating channels, yields a pileup occupancy of only $R\tau \approx 3.3 \times 10^{-5}$, so quiescent reconstruction is not precluded. Every channel is anchored: the deposited CEvNS rate reproduces the Billard germanium benchmark to within 2.4% [7], the muon flux matches the Particle Data Group sea-level value within its $\sim 30\%$ spread, and the Compton edges are fixed exactly by scattering kinematics.

We state the scope plainly. This is a stage-one forward-model feasibility study. It establishes the reconstructed-energy regime in which a QPD germanium wafer would operate; it does *not* compute a sensitivity and makes no claim of detectability or discovery significance. Four modeling uncertainties bound the interpretation and, deliberately, three of them act only on the saturated backgrounds while the flagship result—that reactor CEvNS reconstructs to tens of eV—remains the most robust feature: the unbenchmarked saturated-regime response shape, the factor-of- ~ 2 site-dependent gamma normalization, the cited-but-not-reproduced sub-1.8 MeV reactor flux, and the crossover deposit energy carried as a ~ 50 eV–13 keV band.

Each caveat maps to a concrete next step. A direct QPD saturation calibration would replace the modeled ceiling shape; an on-site gamma survey would fix the absolute Compton normalization; a thin-wafer sensor-sharing measurement would turn the crossover band into a line; and a digitized low-energy reactor flux would collapse the signal-side band. Combined with an exposure model and a noise-and-resolution treatment of the reconstruction floor, these define the path from the present feasibility map to a quantitative QPD reactor-CEvNS sensitivity, extending the phonon-scale reach of the QPD

concept [8] into the reactor-antineutrino program.

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Appendix A: Cross-section and form-factor conventions

This appendix collects the cross-section and form-factor detail deferred from Sec. III: the full Freedman differential cross section with its normalization conventions, the closed-form total against which the differential is checked, the Lewin–Smith Helm form factor and its parameters, and the per-isotope natural-germanium table that enters the fold of Eq. (6).

1. Freedman cross section and closed-form check

For a neutrino of energy E_ν scattering coherently off isotope i with recoil energy $T \equiv E_{\text{nr}}$, we use the Freedman standard-model cross section [1, 2]

$$\frac{d\sigma_i}{dT} = \frac{G_F^2 M_i}{4\pi} Q_{W,i}^2 \left(1 - \frac{M_i T}{2E_\nu^2}\right) F_i^2(q) (\hbar c)^2, \quad (\text{A1})$$

with the Fermi constant G_F , nuclear mass M_i , and weak charge $Q_{W,i} = N_i - (1 - 4\sin^2\theta_W)Z$ evaluated at $\sin^2\theta_W = 0.2387$ (low-energy $\overline{\text{MS}}$ value, not the M_Z value 0.2312) and $Z = 32$. Two normalization choices are load-bearing and are the most common reactor-CEvNS errors: the $/4\pi$ prefactor (the $/8\pi$ form is a factor-of-two error), and the explicit conversion $(\hbar c)^2 = 3.894 \times 10^{-28} \text{ GeV}^2\text{cm}^2$ that restores the natural-unit cross section (evaluated with G_F in GeV^{-2} , M_i and E_ν in GeV) to cm^2 . Because $1 - 4\sin^2\theta_W = 0.0452$ the proton coupling nearly vanishes and $Q_{W,i}^2$ scales as N_i^2 to $\sim 0.5\%$. The kinematic factor $(1 - M_i T/2E_\nu^2)$ enforces the endpoint $T_{\text{max}}^{(i)}(E_\nu) = 2E_\nu^2/(M_i + 2E_\nu)$, and each fold in Eq. (6) runs above $E_\nu^{\text{min},(i)}(T) = \frac{1}{2}(T + \sqrt{T^2 + 2M_i T}) \simeq \sqrt{M_i T/2}$.

Setting $F_i = 1$ and integrating Eq. (A1) over $0 \leq T \leq T_{\text{max}}$ uses $\int_0^{T_{\text{max}}} (1 - M_i T/2E_\nu^2) dT = E_\nu^2/M_i$, giving the closed-form total

$$\sigma_{\text{tot},i}(E_\nu) = \frac{G_F^2 Q_{W,i}^2 E_\nu^2}{4\pi} (\hbar c)^2. \quad (\text{A2})$$

Our differential reproduces Eq. (A2) to a relative 3×10^{-9} – 6×10^{-8} across the five isotopes and $E_\nu \in \{2, 4, 8\} \text{ MeV}$, and yields $\sigma(^{72}\text{Ge}, 4 \text{ MeV}) = 1.0026 \times 10^{-40} \text{ cm}^2$, 0.26% from the first-principles $1.0 \times 10^{-40} \text{ cm}^2$ anchor. Dimensionally $[G_F^2 M_i E_\nu^2] = \text{GeV}^{-1}$, and $(\hbar c)^2$ carries the remaining GeV^2cm^2 , so $d\sigma/dT$ is in cm^2/GeV with $Q_{W,i}^2$, the kinematic factor, and F^2 dimensionless.

TABLE II. The five stable natural-germanium isotopes ($Z = 32$) entering the CEvNS fold of Eq. (6): neutron number N_i , number abundance x_i (IUPAC/CIAAW representative values, $\sum_i x_i = 1$), nuclear mass $M_i = A_i \times 931.494 \text{ MeV}$, and kinematic endpoint $T_{\text{max}}^{(i)}$ at the $E_\nu = 10 \text{ MeV}$ reactor endpoint.

Isotope	N_i	x_i	M_i (GeV)	$T_{\text{max}}^{(i)}$ (eV)
^{70}Ge	38	0.2057	65.13	3066
^{72}Ge	40	0.2745	66.99	2981
^{73}Ge	41	0.0775	67.93	2940
^{74}Ge	42	0.3650	68.86	2901
^{76}Ge	44	0.0773	70.72	2824

2. Lewin–Smith Helm form factor

The nuclear form factor is the Lewin–Smith parametrization [14] of the Helm ansatz [15],

$$F(q) = \frac{3j_1(qR_0)}{qR_0} \exp\left(-\frac{1}{2}q^2 s^2\right), \quad (\text{A3})$$

$$R_0^2 = c^2 + \frac{7}{3}\pi^2 a^2 - 5s^2,$$

with j_1 the first spherical Bessel function, diffraction radius $c = 1.23 A^{1/3} - 0.60 \text{ fm}$, surface diffuseness $a = 0.52 \text{ fm}$, and skin thickness $s = 0.90 \text{ fm}$; $F(0) = 1$ by construction. The momentum transfer $q = \sqrt{2M_i T}$ is expressed in fm^{-1} via $q[\text{fm}^{-1}] = q[\text{MeV}]/(\hbar c)$ with $\hbar c = 197.327 \text{ MeV fm}$, so that qR_0 and qs are dimensionless (feeding q in the wrong units produces a large spurious suppression). For germanium at reactor recoils the form factor is nearly unity: $F^2 > 0.998$ in the sub-100 eV region that carries most of the rate, falling to $F^2(^{72}\text{Ge}) = 0.9963$ at $T = 200 \text{ eV}$ and to 0.9639 only at the rare $\sim 2 \text{ keV}$ endpoint reached by $E_\nu \sim 8$ – 10 MeV neutrinos. These are consistent: the > 0.998 figure is the dominant low-recoil regime and the ~ 0.96 value is the high-recoil tail. The integrated deposited rate above 50 eV is form-factor independent at the sub-percent level—turning F off shifts it by 0.38%—so reactor CEvNS on Ge sits in the form-factor-insensitive regime and the exact skin parameters are not a systematic driver.

3. Natural-germanium isotope table

The fold retains each stable isotope’s own weak charge, nuclear mass, kinematic endpoint, and number abundance rather than a lumped- A approximation, which would smooth the five-step endpoint structure and bias the near-threshold and near-endpoint bins. Table II lists the five isotopes, with $M_i = A_i \times 931.494 \text{ MeV}$ and $T_{\text{max}}^{(i)}$ at the $E_\nu = 10 \text{ MeV}$ reactor endpoint; the per-isotope target density is $N_{T,i} = x_i \times 8.29 \times 10^{24} \text{ kg}^{-1}$ ($\sum_i N_{T,i} = 8.29 \times 10^{24} \text{ kg}^{-1}$). The five distinct endpoints

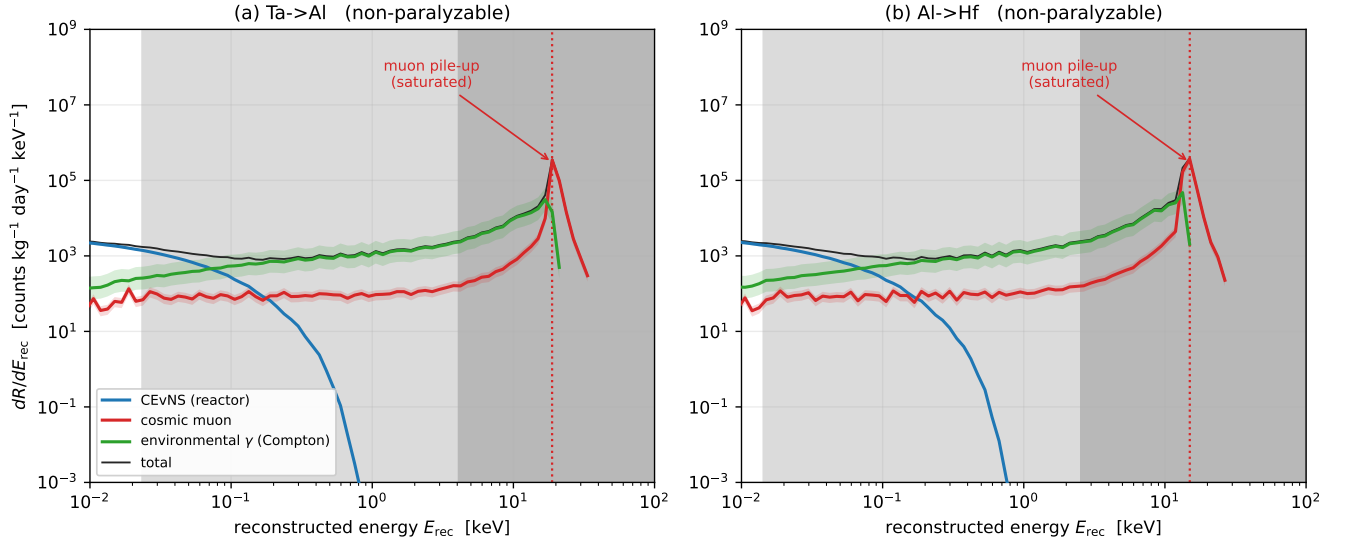


FIG. 1. Reconstructed-energy differential rate spectra dR/dE_{rec} (counts $\text{kg}^{-1} \text{day}^{-1} \text{keV}^{-1}$) for the two QPD trapping designs, (a) Ta→Al and (b) Al→Hf, under non-paralyzable censoring. Reactor CEvNS (blue) reconstructs to tens of eV; cosmic muons (red) pile up at the saturation plateau (tens of keV, an instrumental artifact of the bandwidth ceiling); environmental-gamma Compton (green) fills the continuum between; the total is shown in black. Shaded bands are the propagated flux uncertainties; grey shading marks the saturation region. The underlying true-to-reconstructed energy mapping is shown separately in Fig. 3. Nothing below 10 eV is displayed.

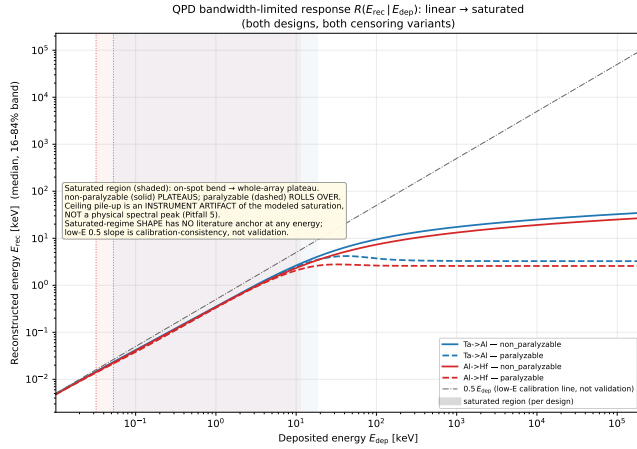


FIG. 2. Monte-Carlo QPD response $R(E_{\text{rec}} | E_{\text{dep}})$ for both designs and both censoring variants, showing the linear $E_{\text{rec}} \approx 0.5 E_{\text{dep}}$ regime, the crossover onset, and the saturation plateau.

(2824–3066 eV) produce the stepped superposition seen at the high-recoil edge of dR/dT .

The ^{73}Ge nuclear spin contributes a spin-dependent axial term that is $\sim 1/N^2$ suppressed and negligible for the coherent rate; we state it but do not model it.

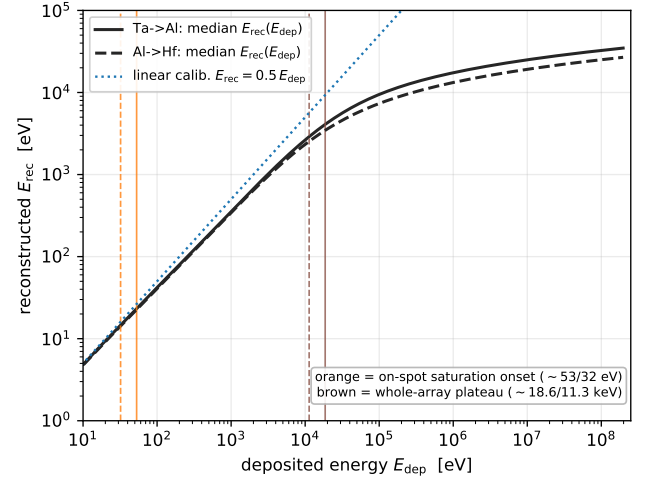


FIG. 3. True-to-reconstructed energy mapping: the median non-paralyzable response $E_{\text{rec}}(E_{\text{dep}})$ for both trapping designs (Ta→Al solid, Al→Hf dashed), with the on-spot saturation-onset (orange) and whole-array-plateau (brown) deposit scales and the linear-calibration reference $E_{\text{rec}} = 0.5 E_{\text{dep}}$ (dotted). Below the onset the mapping is linear; above it the reconstructed energy rolls onto the bandwidth-limited saturation plateau. Nothing below 10 eV is displayed.

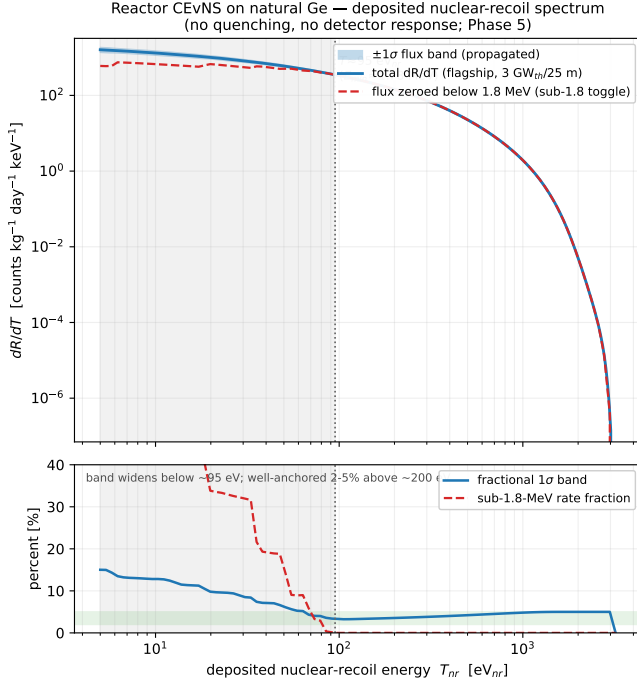


FIG. 4. Deposited nuclear-recoil differential CEvNS rate dR/dT on natural germanium at 3 GW_{th} and 25 m , with the split flux-uncertainty band and the Billard-benchmark reproduction.

Appendix B: Response-matrix Monte Carlo and convergence

This appendix supplies the microphysical detail deferred from Sec. V: the tunneling-burst model that turns a per-sensor share into an event train, the single-constant calibration of the count-integral estimator, the Monte-Carlo construction and convergence of the response matrix $R(E_{\text{rec}} | E_{\text{dep}})$, and the parameter scan that turns the crossover into a band.

EMG tunneling-burst model. On each sensor the trapped-quasiparticle population set by a share E_s drives a two-exponential pulse [Ref. [8], Eq. 4], $N_{\text{qp}}(t) = \tau_{\text{qp}} N_{\text{qp}} g(t)$ with unit-area shape $g(t) = (e^{-t/\tau_{\text{qp}}} - e^{-t/\tau_{\text{inj}}})/(\tau_{\text{qp}} - \tau_{\text{inj}})$, so the instantaneous tunneling rate is $\Gamma_{\text{in}}(t) = (K/V_{\text{tr}})N_{\text{qp}}(t)$. Here τ_{qp} is the trapped-quasiparticle lifetime, τ_{inj} the injection time, K the tunneling-rate constant, and V_{tr} the trap volume. The tunneling events form a burst with exponentially-modified-Gaussian (EMG) arrival times, realized with the `QuasiparticleBurstModel` template of Ref. [8]; its EMG parameters (τ, μ, σ) are fixed once per design so that the pdf peak reproduces $\max_t g(t) = p/\tau_{\text{qp}}$ (recorded peak factors $p = 0.25$ for Al, 0.13 for Hf). The Poisson mean supplied to the template is the expected number of *tunneling events*,

$$n_{\text{ev}} = \int_0^\infty \Gamma_{\text{in}}(t) dt = \frac{K \tau_{\text{qp}}}{V_{\text{tr}}} N_{\text{qp}}, \quad (\text{B1})$$

a dimensionless count that is *not* the trapped population N_{qp} : the two differ by the prefactor $K\tau_{\text{qp}}/V_{\text{tr}} \simeq 0.03$ (Al) and $\simeq 0.008$ (Hf). Feeding N_{qp} instead of n_{ev} would mis-scale the saturation onset by these factors, so Eq. (B1) is enforced everywhere the burst is realized.

Count-integral calibration. Each realized event train is censored by the non-paralyzable dead-time model [Eq. (3)] to a per-sensor observed count, and the array sum N_{obs} is mapped to energy by $E_{\text{rec}} = C N_{\text{obs}}$ [Eq. (15)]. The constant C (eV per event) is a *single global* value per design, fixed once on a deeply unsaturated deposit ($E_{\text{dep}} = 0.01 \text{ eV}$, no sensor near the ceiling) by requiring $dE_{\text{rec}}/dE_{\text{dep}} = \varepsilon = 0.5$. Because the unsaturated summed count is exactly proportional to E_{dep} , one point fixes C ; it is never retuned per bin. The recovery of $E_{\text{rec}} \approx 0.5 E_{\text{dep}}$ at low energy is therefore a calibration-consistency check (verified to $< 1\%$ for both designs), while the deep-saturation plateau is the genuinely predictive content: with censoring disabled the estimator returns exactly $0.5 E_{\text{dep}}$ at all energies, so the plateau is the modeled bandwidth ceiling and not a numerical artifact.

Response-matrix Monte Carlo and convergence. Each E_{dep} column of R is built by histogramming $N_s = 5000$ forward realizations of Eq. (15), on a grid of 584 deposits sampled uniformly in $\log E_{\text{dep}}$ from 10.14 eV to 197 MeV ($\simeq 80$ bins per decade), the shared grid on which Sec. VI folds the deposit spectra. The $\sim \pi r^2$ on-spot sensors are realized individually; the $\sim 10,287$ identical off-spot sensors are drawn as one aggregate multinomial ensemble per realization rather than looped. Columns normalize to unity to 10^{-12} . The per-cell relative Monte-Carlo error is Poisson,

$$\epsilon_{jk} = \frac{1}{\sqrt{N_{jk}}} = \frac{1}{\sqrt{N_s R_{jk}}}, \quad (\text{B2})$$

with N_{jk} the count in cell (j, k) ; the well-populated (peak) cell reaches $\epsilon \lesssim 1.8\%$ at $N_s = 5000$ with clean $1/\sqrt{N_s}$ scaling, and every column is reproducible across processes via deterministic `crc32` sub-seeding. One honesty split is required: where the event count is feasible ($n_{\text{ev}} \leq 2000$) the burst is realized with the full EMG template, but in deep saturation the train would carry $\mathcal{O}(10^8)$ events and is infeasible to materialize, so there the registered count is drawn Poisson about the analytic censored mean. In that regime the relative spread of the summed E_{rec} is $\sim 10^{-3}$ —the column is effectively a delta—so the approximation is immaterial to R ; the per-class mean is pinned to the validated analytic estimator in both branches, leaving no kink at the split.

Crossover-band scan. The on-spot sensor saturates first, at the per-sensor onset share $E_{\text{onset}} (\simeq 1.27 \text{ eV}$ for Ta $\rightarrow$ Al, $\simeq 0.77 \text{ eV}$ for Al $\rightarrow$ Hf); inverting the sharing partition gives the crossover deposit energy $E_{\text{dep}}^{\text{cross}}$ of Eq. (16). Scanning the low-confidence sharing parameters over $f_{\text{prompt}} \in [0.1, 0.5]$ and $r \in [1, 5]$ spreads this from $\sim 8 \text{ eV}$ to $\sim 0.93 \text{ keV}$ (Ta $\rightarrow$ Al) and $\sim 4.8 \text{ eV}$ to $\sim 0.57 \text{ keV}$ (Al $\rightarrow$ Hf) about the default $\sim 52.9/32.1 \text{ eV}$,

the f_{prompt} -dominated band quoted in Sec. V. Two limiting anchors bound it: the equal-split limit $f_{\text{prompt}} \rightarrow 0$, where every sensor shares equally and all saturate together, and the whole-array plateau at the default

f_{prompt} , where even the diffuse share reaches onset,

$$\begin{aligned} E_{\text{dep}}^{\text{equal-split}} &= E_{\text{onset}} N_{\text{sens}}, \\ E_{\text{dep}}^{\text{plateau}} &= \frac{E_{\text{onset}} N_{\text{sens}}}{1 - f_{\text{prompt}}}, \end{aligned} \quad (\text{B3})$$

with $N_{\text{sens}} \simeq 10,300$. These evaluate to ~ 13.1 keV (equal-split) and ~ 18.6 keV (plateau) for Ta $\rightarrow$ Al, and ~ 7.9 keV and ~ 11.3 keV for Al $\rightarrow$ Hf. The Al $\rightarrow$ Hf design, with the lower per-sensor onset, saturates before Ta $\rightarrow$ Al at every point of the scan.

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